

$$\begin{aligned}
 &= \text{Diagram: A vertical line with two 'X' marks at the top and bottom, representing a trace.} \\
 &= \text{Diagram: A square box containing two arcs connecting the top and bottom edges, representing a trace of a product of two operators.} \\
 &\iff \hat{S}[0,1] = \sum_{j,k=0}^{d-1} |j\rangle\langle k| \otimes |k\rangle\langle j|
 \end{aligned}$$